United States Patent	12397595
Kind Code	B2
Date of Patent	August 26, 2025
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Trailer hitch assemblies and methods therefor

Abstract

Trailer hitch assemblies and methods therefor. A trailer hitch assembly includes a mounting connection with a bore configured to receive the elongated stem of a trailer coupling device, in which one or more protuberances on the inner surface of the bore of the mounting connection interact with complementary recesses in the trailer coupling device to prevent the elongated stem from rotating axially within the bore of the mounting connection. A fastener can be inserted through aligned holes through the mounting connection and the elongated stem to secure the trailer coupling device relative to the mounting connection. The mounting connection may be an integral feature of a trailer coupling device, or the mounting connection may be separate from the trailer coupling device.

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Appl. No.: **18/045200**

Filed: **October 10, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230150320 A1	May. 18, 2023

Related U.S. Application Data

us-provisional-application US 63253638 20211008

Publication Classification

Int. Cl.: **B60D1/06** (20060101); **B60D1/52** (20060101)

U.S. Cl.:

CPC **B60D1/06** (20130101); **B60D1/52** (20130101);

Field of Classification Search

CPC: B60D (1/06); B60D (1/065); B60D (1/052)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/253,638, filed Oct. 8, 2021, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION
(1) The present invention generally relates to trailer hitch accessories. The invention particularly relates to trailer hitch assemblies having a mounting connection and a trailer coupling device that cooperate to reduce the likelihood of decoupling during towing while also allowing for more rapid, efficient and safe replacement of the trailer coupling device.
(2) Trailer hitches serve as connectors between a tow vehicle and a trailer. Receiver type trailer hitches generally include a structural component fixed to the tow vehicle chassis that includes a receiver tube that has a rearward facing opening capable of accepting various hitch mounted components therein, such as trailer hitch ball mounts, hitch bike racks, cargo carriers, etc.
(3) FIG. **1** represents a trailer hitch assembly having a common type of trailer hitch mount device (ball mount) **10** that may be releasably coupled to a trailer coupler (hitch) mounted to the tongue of a trailer. The mount device **10** includes a rectangular shank **12** having oppositely-disposed ends and a platform **14** extending from one of the ends. The end of the shank **12** opposite the platform **14** is configured to be inserted into the receiver tube of the trailer hitch such that a pair of holes **16** of the shank **12** align with corresponding holes of the receiver tube of the vehicle. The shank **12** may then be releasably coupled to the receiving tube by inserting a trailer hitch pin **36** through the holes **16**

of the shank **12** and the aligned holes of the receiving tube until a distal end of the trailer hitch pin **36** exits an opposite side of the receiving tube. The trailer hitch pin **36** may then be secured with a fastener such as a cotter pin **38** that may be inserted into a hole **40** of the trailer hitch pin **36**.

(4) The platform **14** provides a mounting connection for a trailer coupling device (trailer ball) **20**. In this example, the mounting connection of the platform is a through hole **18**. The trailer coupling device **20** includes a spherical body (ball) **22** and a stem extending therefrom. The ball **22** can be any one of a variety of sizes as required to receive a particular sized trailer coupler mounted to a tongue of a trailer. The stem includes an upper stem portion **24**, a circumferential flange **26**, and a lower stem portion **28**. The trailer coupling device **20** may be coupled to the mount device **10** by inserting the lower stem portion **28** through the hole **18** of the platform **14** such that the circumferential flange **26** rests on the platform **14** and the lower stem portion **28** extends from an opposite surface of the platform **14** (e.g., the bottom). The lower stem portion **28** can then be secured with a split lock washer **32** and a threaded nut **34** configured to be threadably coupled with threads **30** on the exterior surface of the lower stem portion **28**. Once the mount device **10** is secured to the tow vehicle by adequately torquing the nut **34**, the trailer coupler can be placed on and secured to the ball **22** of the trailer coupling device **20**. This arrangement allows the trailer to swivel and articulate relative to the tow vehicle.

(5) While trailer hitch ball mounts have found continuous and widespread adoption, they are not without shortcomings. For example, during transportation of the trailer, the trailer coupler applies significant radial and rotational forces to the trailer coupling device **20** through its connection with the ball **22**. In addition, the trailer coupling device **20** may experience impact forces and/or vibrations associated with road travel. Over time, these forces may cause the nut **34** and washer **32** to loosen and potentially become decoupled from the lower stem portion **28**. In addition to potentially losing components of the mount device **10**, if the trailer coupling device **20** becomes loose while the tow vehicle is pulling the trailer, for example, on a roadway, significant damage may occur to the tow vehicle, the trailer, and/or other surrounding objects. Consequently, the nut **34** must be torqued to a high torque specification, as examples, about 150 to about 250 ft-lbs. (depending on the size of the lower stem **28**) to inhibit its loosening. As such, current trailer hitch ball mounts on the market often require large wrenches and in many cases a lubricant, considerable force, and several hours to interchange trailer coupling devices **20**, for example, to replace the trailer coupling device **20** with a different trailer coupling device **20** having a larger or smaller ball **22**. In extreme cases, it may be virtually impossible to replace a trailer coupling device **20** from its mounting device **10**.

(6) In view of the above, it can be appreciated that there are certain problems, shortcomings or disadvantages associated with the prior art, and that it would be desirable if a trailer hitch ball mount was available that was capable of at least partly overcoming or avoiding the problems, shortcomings or disadvantages noted above, including the time, efficiency, and simplicity of interchanging trailer balls.

BRIEF SUMMARY OF THE INVENTION

(7) The intent of this section of the specification is to briefly indicate the nature and substance of the invention, as opposed to an exhaustive statement of all subject matter and aspects of the invention. Therefore, while this section identifies subject matter recited in the claims, additional subject matter and aspects relating to the invention are set forth in other sections of the specification, particularly the detailed description, as well as any drawings.

(8) The present invention provides trailer hitch mount devices and methods suitable for securing trailers to tow vehicles.

(9) According to one aspect of the invention, a trailer hitch assembly is provided that includes a trailer hitch mount device have a frame configured to be secured to a tow vehicle, a mounting connection associated with the frame that includes a body with a bore therethrough having a central, longitudinal axis, at least one opening to the bore at an end of the body, a pair of aligned

holes on opposite sides of the body, and one or more protuberances that protrude from interior surfaces of the body into the bore and extend within the bore in a direction parallel to the longitudinal axis of the bore, and a trailer coupling device configured to be secured to a trailer that includes an elongated stem having one or more recesses formed in exterior surfaces thereof that extend along the elongated stem in a direction parallel to the longitudinal axis of the elongated stem, and a hole through a diameter of the elongated stem having openings on opposite sides thereof. The elongated stem is configured to be received within the bore of the mounting connection such that the one or more protuberances of the mounting connection are aligned with and received within the one or more recesses of the elongated stem and the holes of the mounting connection are aligned with the hole in the stem. The trailer hitch mount device includes a fastener configured to be inserted into a first of the holes of the mounting connection, through the hole of the elongated stem, and out of a second of the holes of the mounting connection while the elongated stem is received in the bore to secure the trailer coupling device relative to the mounting connection in a direction parallel to the longitudinal axis of the bore. The one or more protuberances of the mounting connection act as barriers on the recesses of the elongated stem and thereby prevent the elongated stem from rotating within the bore of the mounting connection about the longitudinal axis thereof.
(10) According to another aspect of the invention, a method is provided for coupling a trailer to a tow vehicle. The method includes securing a frame of a trailer hitch mount device to the tow vehicle, aligning a central, longitudinal axis of an elongated stem of a trailer coupling device with a central, longitudinal axis of a bore of a mounting connection associated with the frame, aligning one or more protuberances that protrude from interior surfaces of the mounting connection into the bore with one or more recesses formed in exterior surfaces of the elongated stem, inserting the elongated stem of the trailer coupling device into an opening to the bore of the mounting connection such that the one or more protuberances of the mounting connection are received within the one or more recesses of the elongated stem, continuing to insert the elongated stem into the bore until a hole through a diameter of the elongated stem having openings on opposite sides thereof is aligned with a pair of aligned holes on opposite sides of the mounting connection, inserting a fastener into a first of the holes of the mounting connection, through the hole of the elongated stem, and out of a second of the holes of the mounting connection, coupling a trailer to the trailer coupling device, and restricting rotation of the elongated stem within the bore by contacting surfaces of the elongated stem that define the one or more recesses with the one or more protuberances of the mounting connection.
(11) Technical effects of trailer hitch assemblies, trailer hitch mount devices, and methods described herein may include the ability to secure the trailer coupling device to the frame of the trailer hitch mount device without using threaded connections that may become decoupled during use.
(12) These and other aspects, arrangements, features, and/or technical effects will become apparent upon detailed inspection of the figures and the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** contains perspective views of components of a trailer hitch assembly with a conventional trailer hitch mount device for securing a trailer to a tow vehicle.
(2) FIG. **2** is a perspective view of a trailer hitch assembly with a trailer hitch mount device for securing a trailer to a tow vehicle in accordance with a first nonlimiting embodiment of the invention.
(3) FIG. **3** is an enlarged perspective view of a mounting connection of the mount device of FIG. **2**.

(4) FIG. **4** is a bottom end view of a cylindrical body of the mounting connection of FIG. **3**.
(5) FIG. **5** is a perspective view of a trailer coupling device suitable for use with the mount device of FIG. **2** in accordance with certain nonlimiting aspects of the invention.
(6) FIG. **6** is a bottom end view of the trailer coupling device of FIG. **5**.
(7) FIG. **7** is a transverse cross-sectional view of a lower stem portion of the trailer coupling device generally along lines **7-7** of FIG. **5**.
(8) FIG. **8** is an enlarged partial cross-sectional view of a trailer hitch assembly with an alternative mounting connection to the mounting connection of FIGS. **2** through **4** in accordance with an additional nonlimiting embodiment of the invention.
(9) FIG. **9** is a side elevation view of the trailer hitch assembly of FIG. **8** with the mounting connection operatively assembled with the trailer coupling device of FIG. **5** and the trailer hitch mount device of FIG. **1**.
(10) FIG. **10** is an isometric view of the trailer coupling device of FIG. **5** disposed in the mounting connection of FIG. **8** with the mounting connection shown in phantom for ease of seeing otherwise hidden features.
(11) FIG. **11** is a perspective view of another alternative mounting connection to the mounting connection of FIGS. **2** through **4** in accordance with yet an additional nonlimiting embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(12) The intended purpose of the following detailed description of the invention and the phraseology and terminology employed therein is to describe what is shown in the drawings, which include the depiction of one or more nonlimiting embodiments of the invention, and to describe certain but not all aspects of what is depicted in the drawings, including the embodiment(s) depicted in the drawings. The following detailed description also identifies certain but not all alternatives of the embodiment(s) depicted in the drawings. As nonlimiting examples, the invention encompasses additional or alternative embodiments in which one or more features or aspects shown and/or described as part of a particular embodiment could be eliminated, and also encompasses additional or alternative embodiments that combine two or more features or aspects shown and/or described as part of different embodiments. Therefore, the appended claims, and not the detailed description, are intended to particularly point out subject matter regarded to be aspects of the invention, including certain but not necessarily all of the aspects and alternatives described in the detailed description.
(13) To facilitate the description provided below of the embodiment(s) represented in the drawings, relative terms, including but not limited to, “proximal,” “distal,” “anterior,” “posterior,” “vertical,” “horizontal,” “lateral,” “front,” “rear,” “side,” “forward,” “rearward,” “top,” “bottom,” “upper,” “lower,” “above,” “below,” “right,” “left,” etc., may be used in reference to the orientation of the mounting connections and coupling devices during use and/or as represented in the drawings. All such relative terms are useful to describe the illustrated embodiment(s) but should not be otherwise interpreted as limiting the scope of the invention.
(14) Disclosed herein are trailer hitch assemblies including mounting connections **118** and trailer coupling devices (e.g., trailer balls) **120** that are adapted for use in combination with trailer hitch mount devices (ball mounts), including but not limited to the mount device **10** of FIG. **1**, for coupling trailers to tow vehicles. The mounting connections **118** and trailer coupling devices **120** replace the threaded connections of certain conventional trailer hitch mount devices, such as that of the conventional mount device **10** represented in FIG. **1**, which may loosen and/or become lost during transportation of the trailer with the tow vehicle. The mounting connections **118** and trailer coupling devices **120** includes one or more complementary features, represented as protuberances **146** and corresponding recesses(s) **152**, that interlock to restrict rotational motion of a connection point between the trailer and the tow vehicle. The mounting connections **118** and trailer coupling devices **120** may also include a pin-type fastener for restricting their vertical motion relative to the

trailer and tow vehicle.
(15) FIGS. **2** through **4** illustrate a trailer hitch assembly including a first embodiment of the mounting connections **118**, FIGS. **8** through **11** illustrate a trailer hitch assembly with alternative embodiments of the mounting connections **118**, and FIGS. **5** through **7** illustrate the trailer coupling device **120** adapted for use with trailer hitch assemblies including each of the illustrated embodiments of mounting connections **118**. Referring initially to FIGS. **2** through **4**, a trailer hitch mount device **110** is shown that is similar to the mount device **10** of FIG. **1** except for integrally incorporating the mounting connection **118** of the first embodiment. The mount device **110** includes a frame having a shank **112** configured to be coupled to a receiver type trailer hitch. The shank **112** has oppositely-disposed ends, with a platform **114** extending from one of the ends. The end of the shank **112** opposite the platform **114** is configured to be inserted into the receiver tube of a trailer hitch such that a pair of holes **116** of the shank **112** can be aligned with corresponding holes of the receiver tube. The shank **112** may then be releasably coupled to the receiving tube by inserting a trailer hitch pin **136** through the holes **116** of the shank **112** and the aligned holes of the receiving tube until a distal end of the trailer hitch pin **136** exits an opposite side of the receiving tube. The trailer hitch pin **136** may then be secured with a fastener such as a cotter pin **138** that may be inserted into a hole **140** of the trailer hitch pin **136**.
(16) In FIG. **2**, the mounting connection **118** includes an opening **144**A in the upper surface of the platform **114** and a cylindrical body **144**B extending downward from the lower surface of the platform **114**. In this embodiment, the cylindrical body **144**B is fixed to the platform **114** in any suitable manner. Together, the opening **144**A and cylindrical body **144**B define an elongated cavity (bore) **144**C (FIG. **4**) that extends entirely through the platform **114** and cylindrical body **144**B, from the upper surface of the platform **114** and through the lower end of the cylindrical body **144**B. The opening **144**A and bore **144**C are represented as sharing a central, longitudinal axis.
(17) The cylindrical body **144**B in FIGS. **2** and **3** is an integral feature of the platform **114** as a result of the cylindrical body **144**B being machined, cast, or otherwise formed as part of the platform **114**. Alternatively, the cylindrical body **144**B could be separately fabricated and then assembled with or attached to the platform **14** of FIG. **1** or a modification thereof, as discussed below in reference to FIGS. **8** through **11**.
(18) In FIG. **2**, the cylindrical body **144**B is centrally located in the platform **114** in relation to the width of the platform **14** and located adjacent a distal end of the platform **114** relative to the shank **112**, though it is foreseeable that the cylindrical body **144**B could be attached to the distal end or to a side of the platform **114**. Also, though the cylindrical body **144**B in this example extends only from the lower surface of the platform **114**, the cylindrical body **144**B could extend through the platform **114** such that portions of the cylindrical body **144**B extend from one or both upper and lower surfaces of the platform **114**.
(19) A pair of protuberances **146** extend from interior surfaces of the opening **144**A and cylindrical body **144**B to project radially inward into the bore **144**C. In this nonlimiting embodiment, the protuberances **146** define continuous elongated rails or ribs, each having a V-shaped cross-section and extending along the entire length of the bore **144**C within the body **144**B and in a direction parallel to the longitudinal axis of the bore **144**C. FIG. **4** illustrates in phantom an alternative shape of the protuberances **146** as being arcuate in shape. Further alternatives include the mounting connection **118** having fewer or more protuberances **146**, protuberances **146** that are not continuous along their lengths or otherwise do not extend along an entire length of the bore **144**C, and/or protuberances **146** that have different cross-sectional shapes from each other.
(20) As best seen in FIGS. **3** and **4**, a single pair of aligned holes **148** are formed on opposite sides of the cylindrical body **144**B. Although not required, in this embodiment the holes **148** have a common central axis that is perpendicular to a geometric line between the pair of protuberances **146**. In the embodiment of FIG. **11**, more than one pair of aligned holes **148** are provided, one pair above the other, so that the length of the cylindrical body **144**B can be sized to accommodate

platforms **14** or **114** of different thicknesses, for example, a 1-inch thick hitch on a dump truck or a ½-inch thick hitch on a smaller vehicle, to promote the universality of the mounting connection **118**.

(21) Referring now to FIGS. **5** through **7**, the trailer coupling device **120** (e.g., a trailer ball) is represented as configured to be secured to the mounting connection **118** of the platform **114** and function as the connection between the mounting device **110** and a trailer. The trailer coupling device **120** includes a spherical body (ball) **122** configured to mate with a trailer coupler on a trailer tongue of a trailer. As with the ball **22** of FIG. **1**, the ball **122** can be any one of a variety of sizes that may be required to receive a particular trailer coupler mounted to a trailer tongue. In the nonlimiting embodiment shown, an elongated stem extending from the ball **122** includes an upper stem portion **124** adjacent the ball **122**, a circumferential flange **126** adjacent the upper stem portion **124**, and a lower stem portion **128** extending from the circumferential flange **126**.

(22) The lower stem portion **128** is sized and configured to be received within the bore **144**C of the mounting connection **118** to secure the trailer coupling device **120** to the mounting device **110**. Therefore, since in this embodiment the bore **144**C is cylindrical except for the protuberances **146**, the lower stem portion **128** of the trailer coupling device **120** may also be cylindrical with the exception of recesses **152** formed in exterior surfaces thereof and configured to receive the protuberances **146** of the mounting connection **118**. The cross-sectional shapes of the recesses **152** are preferably complementary to the protuberances **146**, in this case, as a result of defining valleys having V shaped cross sections extending along the exterior surfaces of the lower stem portion **128** parallel to a central, longitudinal axis of the elongated stem. Alternatively, the lower stem portion **128** may include fewer or more recesses **152**, may have recesses **152** that do not extend along an entirety of the bore **144**C, may have recesses **152** having other cross-sectional shapes, and/or may have recesses **152** with different cross-sectional shapes. The sizes and shapes of the recesses **152** are only limited by the size and shapes of the corresponding protuberances **146** of the mounting connection **118** that must be receivable within the recesses **152**.

(23) The lower stem portion **128** may also include a hole **154** through a diameter thereof having openings on opposite sides thereof. In certain embodiments, the lower stem portion **128** is configured to be located within the bore **144**C to an extent such that the circumferential flange **126** rests on the platform **114**. In such embodiments, inserting the lower stem portion **128** through the bore **144**C to the extent that the circumferential flange **126** contacts the platform **114** causes the hole **154** of the lower stem portion **128** and the pair of holes **148** of the mounting connection **118** to be axially aligned along a single axis.

(24) In the arrangement of FIGS. **3** through **7**, the central, longitudinal axes of the holes **148** of the mounting connection **118** are perpendicular to a geometric line or plane extending between the two protuberances **146**, and a central, longitudinal axis of the hole **154** of the lower stem portion **128** is perpendicular to a geometric line or plane extending between the two recesses **152**. It should be understood that this arrangement is merely exemplary, and the relative positions of the holes **148** of the mounting connection **118** and the protuberances **146**, and the hole **154** of the lower stem portion **128** and the recesses **152** may have other arrangements. However, regardless of the specific arrangement between these elements, the holes **148** of the mounting connection **118** and the hole **154** of the lower stem portion **128** should be capable of aligning when the lower stem portion **128** is received within the bore **144**C and the protuberances **146** are received within the recesses **152**.

(25) Once the lower stem portion **128** has been received within the bore **144**C and the holes **148** of the mounting connection **118** and the hole **154** of the lower stem portion **128** are aligned, a fastener may be inserted into one of the holes **148** of the mounting connection **118**, through the hole **154** of the elongated stem, and out through the other paired hole **148** of the mounting connection **118**. For example, FIG. **5** represents the trailer coupling device **120** as including a trailer ball pin **156** (e.g., a trailer hitch pin or miniature trailer hitch pin) configured to be received within the holes **148** of the mounting connection **118** and the hole **154** of the lower stem portion **128** and secured by inserting a

secondary fastener, such as a cotter pin **158**, through a hole **160** in the trailer ball pin **156**.
(26) With the trailer coupling device **120** assembled with the mounting connection **118** as described above, the protuberances **146** of the mounting connection **118** may contact surfaces of the lower stem portion **128** that define the recesses **152** and act as barriers that restrict if not prevent rotation of the trailer coupling device **120** within the bore **144**C of the mounting connection **118** about the longitudinal axis thereof. In addition, the trailer ball pin **156** may contact edges of the holes **148** of the mounting connection **118** and act as a barrier that restricts movement of the trailer coupling device **120** within the bore **144**C in a direction parallel to the longitudinal axis of the bore **144**C.
(27) FIGS. **8** through **10** illustrate another example embodiment of a trailer hitch assembly, in which the mounting connection **118** and its cylindrical body **144**B may be separately fabricated and then assembled with the platform **14**, as previously noted. In this embodiment, the mounting connection **118** is formed essentially of only the cylindrical body **144**B. Unlike the embodiment of FIGS. **2** through **4**, the mounting connection **118** in this embodiment is separate from and not fixedly attached to the platform **14**, that is, it is not integrally incorporated as a part of the platform **14**. FIGS. **8** and **9** illustrate the mounting connection **118** of this embodiment operatively assembled with the platform **14** of the conventional mount device **10** of FIG. **1** by aligning the bore **144**C of the cylindrical body **144**B with the hole **18** in the platform **14** and abutting the upper end of the cylindrical body **144**B (in which the opening **144**A is formed) against the lower surface **14**A of the platform **14**, and then securing the cylindrical body **144**A to the platform **14** with the trailer coupling device **120**. In this embodiment, no modifications are required of the mount device **10** are required and the mounting connection **118** and trailer coupling device **120** may be able to rotate in unison within the hole **18** in the platform **14**. Alternatively, the cylindrical body **144**B could be fixedly attached to the lower surface **14**A of the platform **114** beneath the opening **18**, for example with a weld or other type of permanent coupling. In this arrangement, the hole **18** through the platform **14** does not include the protuberances **146**, but rather has a substantially smooth inner surface, such as that of a substantially cylindrical through bore.
(28) As best seen in FIGS. **8** and **10**, the mounting connection **118** has a generally cylindrical body **144**B with the elongated cavity **144**C formed by a central through bore extending along the central axis of the cylindrical body **144**B. Two radially inwardly projecting protuberances **146** in the form of V-shaped vertical ribs extending the length of the elongated cavity **144**C are disposed on diametrically opposite sides of the inner surface of the elongated cavity **144**C. A pair of holes **148** through opposite sides of the circumferential wall of the cylindrical body **144**B are axially diametrically aligned to receive the pin **156** therethrough. The holes **148** are disposed axially along the length of the cylindrical body **144**B approximately midway between the top end and the bottom end of the cylindrical body **144**B, although other locations are possible. In this arrangement, the axis through the holes **148** is orthogonal to a plane defined by the two protuberances **146**.
(29) On the trailer coupling device, the recesses **152** are diametrically aligned on opposite sides of the lower stem portion **128**, and the hole **154** is axially aligned along a diameter that is orthogonal to a plane define by the two recesses **152**. In this way, the sliding interfit of the protuberances **146** into the recesses **152** helps ensure proper alignment of the holes **148** of the mounting connection **118** with the hole **154** of the trailer coupling device **120** when assembling the two pieces together, thereby eliminating or reducing the difficulty of aligning all three holes so that the pin **156** can be inserted therethrough. This can make it easier and faster to operatively assembly the mounting connection **118** to the trailer coupling device **120**.
(30) As best seen in FIG. **10**, the outer surface of the cylindrical body **144**B has a flattened portion **148**D, the outer periphery of the circumferential flange **126** has a flattened surface **126**A that are generally aligned with each other, for example along a common plane or parallel planes, when the hole **154** is axially aligned with the holes **148**. The flattened portion **148**D is generally orthogonal to the hole **148**, and the plane of the flattened portion **126**A is generally orthogonal to the hole **154**. Optionally, a second set of complementary flattened portions **126**A and **148**D may also be disposed

on the opposite sides of the respective circumferential flange **126** and cylindrical portion **144**B, the back side as seen in FIG. **10**. These flattened portions **126**A and **148**D can provide an additional aid to help a user rotationally align the mounting connection **118** with the trailer coupling device **120** when assembling them together by visually and/or tactilely aligning the flattened portions **126**A and **148**D when bringing the mounting connection and the trailer coupling device together.

(31) As best seen in FIG. **9**, to assemble the trailer hitch assembly with the trailer coupling device **120** in an operative configuration on the trailer hitch mount device **10**, the lower stem portion **128** of the trailer coupling device **120** slides (downwardly) into and through the hole **18** through the platform **14** of the mount device **10**. Next, the mounting connection **118** slides (upwardly) onto the portion of the lower stem portion **128** extending below the bottom surface of the platform. To do this, the lower stem portion **128** is inserted into the opening **144**A of the mounting connection **118** with the protuberances **146** in the through bore **144**A axially aligned with the recesses **152** on the lower stem portion **128** until the holes **148** are axially aligned with the hole **154**. With the holes **148** and **154** so aligned, the pin **156** is inserted through the aligned holes **148** and **154**, thereby locking the mounting connection **118** to the lower stem portion **144**B of the trailer coupling device **120**, which also has the effect of operatively assembling, mounting, and securing the trailer coupling device **120** to the trailer hitch mount device **10**. As before, the pin **156** can further be locked into place extending through the holes **148** and **154** with a cotter pin **158** or other secondary fastener. If it later desired to remove the trailer coupling device **120** from the trailer hitch mount device **10**, for example to change out to a different hitch or have no ball hitch, this process can simply be reversed. Thus, the example embodiment of FIGS. **8** through **11** provides an embodiment of the invention that can be easily retrofitted to and operatively used with the conventional trailer hitch mount device **10** without having to permanently alter the trailer hitch mount device by permanently affixing the mounting connection **118** thereto.

(32) FIG. **11** shows a portion of another alternative of a trailer hitch assembly in which the mounting connection **118** may be placed in the existing hole **18** in the platform **14** of FIG. **1** or in an enlarged hole in a modified platform. In the nonlimiting example of FIG. **11**, the cylindrical body **144**B is separately fabricated and received in the hole **18** (or enlarged hole) in the platform **14**, with the upper end of the cylindrical body **144**B having a flange **144**D that rests on the upper surface of the platform **14**. In the example of FIG. **11**, the flange **144**D could be omitted, the cylindrical body **144**B could be secured within the hole **18** (or enlarged hole) by welding, and/or the cylindrical body **144**B and hole **18** (or enlarged hole) could have complementary threads such that the cylindrical body **144**B is secured to the platform **14** with threads alone or in combination with a weld. Alternatively, the cylindrical body **144**B could be welded to the platform **14** beneath the opening **18**.

(33) The mounting connections **118** described above provide for methods of coupling a trailer to a tow vehicle. The method may include securing the frame of the trailer hitch mount device **110** to the tow vehicle, for example, by inserting the shank **112** into the receiver tube of the trailer hitch such that the pair of holes **116** of the shank **112** align with the corresponding holes of the receiver tube of the tow vehicle. The shank **112** may then be secured with the trailer hitch pin **136**.

(34) To couple the trailer coupling device **120**, the method may include aligning the central, longitudinal axis of the elongated stem of the trailer coupling device **120** with the central, longitudinal axis of the bore **144**C of the mounting connection **118**, and simultaneously aligning the one or more protuberances **146** extending from the interior surfaces of the mounting connection **118** into the bore **144**C with the recesses **152** formed in the exterior surfaces of the elongated stem. Once all of these elements are aligned, the lower stem portion **128** of the trailer coupling device **120** may be inserted into the opening to the bore **144**C of the mounting connection **118** such that the protuberances **146** are received within the recesses **152**, and to an extent that the hole **154** through the elongated stem is aligned with the pair of holes **148** of the mounting connection **118**.

(35) In the embodiments represented in the drawings, this may include inserting the lower stem

portion **128** of the elongated stem into the bore **144**C of the mounting connection **118** until the circumferential flange **126** rests on the frame which causes the hole **154** of the lower step portion and the holes **148** of the mounting connection **118** to be aligned. The trailer coupling device **120** may then be secured with a fastener such as the trailer ball pin **156**. Thereafter, the trailer may be secured to the trailer coupling device **120**. Generally, this includes covering the ball **122** of the trailer coupling device **120** with the trailer coupler and locking the trailer coupler thereon. As the trailer is towed with the tow vehicle, rotation of the elongated stem is preferably restricted within the bore **144**C as a result of surfaces of the recesses **152** formed in the lower stem portion **128** contacting surfaces of the protuberances **146** of the mounting connection **118**.

(36) While the invention has been described in terms of a specific or particular embodiment, it should be apparent that alternatives could be adopted by one skilled in the art. For example, the trailer hitch mount device **110** and its components could differ in appearance and construction from the embodiment described herein and shown in the figures, functions of certain components of the trailer hitch mount device **110** could be performed by components of different construction but capable of a similar (though not necessarily equivalent) function, and various materials could be used in the fabrication of the trailer hitch mount device **110** and/or its components. Accordingly, it should be understood that the invention is not necessarily limited to any embodiment described herein.

Claims

1. A trailer hitch assembly, comprising: trailer hitch mount device comprising a frame configured to be secured to a tow vehicle; a mounting connection associated with the frame that includes a body with a bore therethrough having a central, longitudinal axis, at least one opening to the bore at an upper end of the body, a pair of aligned holes on opposite sides of the body, and one or more protuberances that protrude from interior surfaces of the body into the bore and extend within the bore in a direction parallel to the longitudinal axis of the bore; and a trailer coupling device configured to be secured to a trailer that includes an elongated stem having one or more recesses formed in exterior surfaces thereof that extend along the elongated stem in a direction parallel to the longitudinal axis of the elongated stem, and a hole through a diameter of the elongated stem having openings on opposite sides thereof; wherein the elongated stem is configured to be received within the bore of the mounting connection such that the one or more protuberances of the mounting connection are aligned with and received within the one or more recesses of the elongated stem and the holes of the mounting connection are aligned with the hole in the elongated stem; a fastener configured to be inserted into a first of the holes of the mounting connection, through the hole of the elongated stem, and out of a second of the holes of the mounting connection while the elongated stem is received in the bore to secure the trailer coupling device relative to the mounting connection in a direction parallel to the longitudinal axis of the bore; wherein the one or more protuberances of the mounting connection act as barriers on the recesses of the elongated stem and thereby prevent the elongated stem from rotating within the bore of the mounting connection about the longitudinal axis thereof.
2. The trailer hitch assembly of claim 1, wherein the bore is cylindrical except for the one or more protuberances and the stem of the trailer coupling devices is cylindrical except for the one or more recesses.
3. The trailer hitch assembly of claim 1, wherein the frame comprises: a shank configured to be coupled to a receiver-type trailer hitch; and a platform extending from the shank that includes the mounting connection.
4. The trailer hitch assembly of claim 1, wherein the trailer hitch mount device is a trailer hitch ball mount and the trailer coupling device is a trailer ball that includes a spherical body configured to mate with a trailer coupler.

5. The trailer hitch assembly of claim 4, wherein the elongated stem extends from the spherical body and comprises: an upper stem portion adjacent the spherical body; a circumferential flange adjacent the upper stem portion; and a lower stem portion extending from the circumferential flange; wherein the lower stem portion includes the one or more recesses and the hole therethrough; and wherein the lower stem portion is configured to be located within the bore to an extent such that the circumferential flange rests on the platform.
6. The trailer hitch assembly of claim 1, wherein the fastener is a trailer hitch pin comprising a secondary fastener configured to secure the fastener within the hole of the elongated stem and the holes of the mounting connection.
7. The trailer hitch assembly of claim 1, wherein the one or more protuberances define elongated ribs having V-shaped cross-sections and the one or more recesses device elongated valleys having V-shaped cross-sections.
8. The trailer hitch assembly of claim 1, wherein the mounting connection includes two of the one or more protuberances on opposite sides of the bore and the trailer coupling device includes two of the one or more recesses on opposite sides of the elongated stem, and wherein inserting the elongated stem into the bore such that the two protuberances mate with the two recesses causes the hole of the elongated stem and the holes of the mounting connection to be aligned.
9. The trailer hitch assembly of claim 8, wherein a central, longitudinal axis of the hole of the elongated stem is perpendicular to a geometric line or plane extending between the two recesses.
10. The trailer hitch assembly of claim 1, wherein the body of the mounting connection is an integral feature of the frame.
11. The trailer hitch assembly of claim 1, wherein the body of the mounting connection is separately fabricated from the frame and aligned with a through-hole in the frame.
12. The trailer hitch assembly of claim 11, wherein the upper end of the cylindrical body is abutted against a lower surface of the frame and secured to the frame with the trailer coupling device.
13. A method for coupling a trailer to a tow vehicle, the method comprising: securing a frame of a trailer hitch mount device to the tow vehicle; aligning a central, longitudinal axis of an elongated stem of a trailer coupling device with a central, longitudinal axis of a bore of a mounting connection associated with the frame; aligning one or more protuberances extending from interior surfaces of the mounting connection into the bore with one or more recesses formed in exterior surfaces of the elongated stem; inserting the elongated stem of the trailer coupling device into an opening to the bore of the mounting connection such that the one or more protuberances of the mounting connection are received within the one or more recesses of the elongated stem; continuing to insert the elongated stem into the bore until a hole through a diameter of the elongated stem having openings on opposite sides thereof is aligned with a pair of aligned holes on opposite sides of the mounting connection; inserting a fastener into a first of the holes of the mounting connection, through the hole of the elongated stem, and out of a second of the holes of the mounting connection; coupling a trailer to the trailer coupling device; and restricting rotation of the elongated stem within the bore by contacting surfaces of the elongated stem that define the one or more recesses with the one or more protuberances of the mounting connection.
14. The method of claim 13, wherein securing the frame of the trailer hitch mount device to the tow vehicle comprises coupling a shank of the frame to a receiver-type trailer hitch of the tow vehicle.
15. The method of claim 13, wherein the trailer hitch mount device is a trailer hitch ball mount and the trailer coupling device is a trailer ball that includes a spherical body, wherein coupling the trailer to the trailer coupling device includes coupling a trailer coupler to the spherical body.
16. The method of claim 13, wherein the elongated stem extends from the spherical body and comprises an upper stem portion adjacent the spherical body, a circumferential flange adjacent the upper stem portion, and a lower stem portion extending from the circumferential flange, wherein the elongated stem is inserted into the bore until the circumferential flange rests on the frame.
17. The method of claim 13, wherein the fastener is a trailer hitch pin, the method further

comprising securing the trailer hitch pin with a secondary fastener.
18. The method of claim 13, wherein the one or more protuberances define elongated ribs having V-shaped cross-sections and the one or more recesses device elongated valleys having V-shaped cross-sections, wherein the one or more protuberances are received within and slide through the one or more recesses when the elongated stem is inserted into the bore.
19. The method of claim 13, wherein the mounting connection includes two of the one or more protuberances on opposite sides of the bore and the trailer coupling device includes two of the one or more recesses on opposite sides of the elongated stem, wherein inserting the elongated stem into the bore such that the two protuberances mate with the two recesses causes the hole of the elongated stem and the holes of the mounting connection to be aligned.
20. The method of claim 13, wherein the body of the mounting connection is separately fabricated from the frame and aligned with a through-hole in the frame, the method further comprising: abutting an upper end of the cylindrical body against a lower surface of the frame; and securing the cylindrical body to the frame with the trailer coupling device.